Further Mathematics GCE A Level 2024 - Paper 2

Duration: 3h | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. Further Mathematics Paper 2 (0775), ADVANCED LEVEL, June 2024. Duration: three hours.

Answer ALL 10 questions. For your guidance, the approximate mark allocation for parts of each question is indicated. Mathematical formulae and tables published by the Board, and noiseless non-programmable electronic calculators are allowed. In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Exercise 1

i. The differential equation

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 4e^{-3x}$$

has general solution $Ae^{-3x} + Bxe^{-3x} + p(x)$, where $p(x)$ is its particular integral.

Find $p(x)$.

ii. Given that $y = 1$ and $\frac{dy}{dx} = 1$ when $x = 0$, find the values of A and B.

Exercise 2

i. Given that a group G has identity element e and that $a^3 = a$ for all $a \in G$, show that $a^2 = e$.

ii. Construct the operation table for $H = (\{1, 5, 7, 11\}, \times_{12})$, where $\times_{12}$ is multiplication modulo 12.

iii. Hence or otherwise solve completely the equation $x^3 = x$.

iv. Hence or otherwise solve the system of congruences

$$5x \equiv 2 \pmod{12}, \qquad x \equiv 2 \pmod{7}.$$

Exercise 3

a. Express $f(x) = \dfrac{2}{(x+1)^2(x^2+1)}$, $x \neq -1$, in partial fractions.

b(i). The lines L_1 and L_2 are defined by

$$L_1 : \ \mathbf{r} = -4\mathbf{i} + 4\mathbf{j} + 7\mathbf{k} + t(2\mathbf{i} - \mathbf{j} + \mathbf{k}),$$

$$L_2 : \ \mathbf{r} = -2\mathbf{i} + \mathbf{j} + 6\mathbf{k} + s(2\mathbf{i} + \mathbf{j} + \mathbf{k}), \qquad t, s \in \mathbb{R}.$$

Find the coordinates of the foot of the perpendicular from the point $P(2, -1, 8)$ to L_1.

b(ii). The lines L_1 and L_2 are defined by

$$L_1 : \ \mathbf{r} = -4\mathbf{i} + 4\mathbf{j} + 7\mathbf{k} + t(2\mathbf{i} - \mathbf{j} + \mathbf{k}),$$

$$L_2 : \ \mathbf{r} = -2\mathbf{i} + \mathbf{j} + 6\mathbf{k} + s(2\mathbf{i} + \mathbf{j} + \mathbf{k}), \qquad t, s \in \mathbb{R}.$$

Find the Cartesian equation of the plane containing L_1 and L_2.

Exercise 4

i. A curve has equation $y = 4\sqrt{x}$, $x > 0$.

Show that the length of the arc, S, of the curve for which $1 \leq x \leq 4$ is given by

$$S = \int_1^4 \sqrt{\frac{x+4}{x}}\, dx.$$

ii. A curve has equation $y = 4\sqrt{x}$, $x > 0$, and the length of its arc for $1 \leq x \leq 4$ is $S = \displaystyle\int_1^4 \sqrt{\frac{x+4}{x}}\, dx$.

Using the substitution $x = 4\sinh^2\theta$, find S (leaving your answer in logarithmic form).

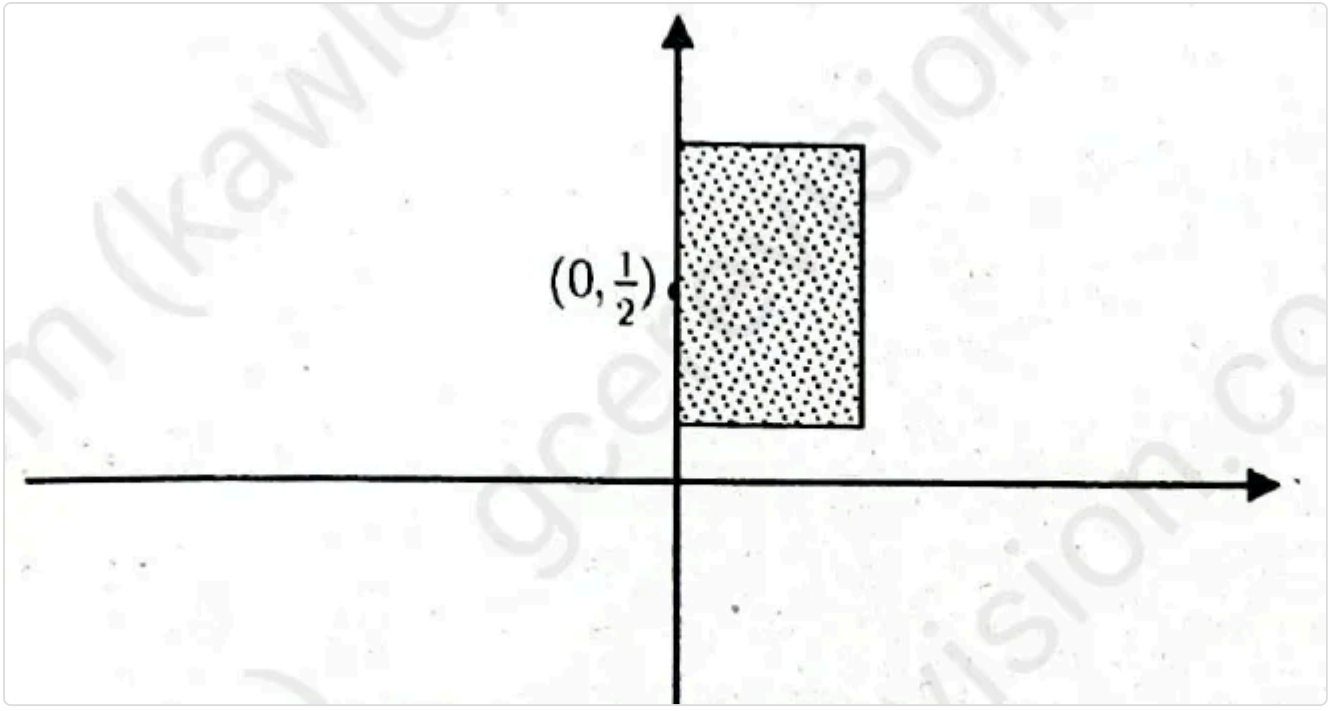
Exercise 5

a(i). A similarity transformation (similitude) is defined by $z' = 3z - 2i$.

Show that in the complex plane, the invariant point of the transformation is $\left(0, \frac{1}{2}\right)$.

a(ii). A similarity transformation (similitude) is defined by $z' = 3z - 2i$.

Given the object z in the diagram below, copy it, and on the same axes sketch its image, z'.



b. Show that the point with polar coordinates (r, θ) is also the point with Cartesian coordinates whose position vector is $\mathbf{r} = r\cos\theta\,\mathbf{i} + r\sin\theta\,\mathbf{j}$.

b(i). Two points A and B lie on a plane which contains the origin O. The position vector of A is $\mathbf{r} = -\mathbf{i} + \sqrt{3}\,\mathbf{j}$, while the polar coordinates (r, θ) of B are $\left(2, \frac{1}{6}\pi\right)$.

Find the position vector of B and the polar coordinates of A.

b(ii). Two points A and B lie on a plane which contains the origin O. The position vector of A is $\mathbf{r} = -\mathbf{i} + \sqrt{3}\,\mathbf{j}$, while the polar coordinates (r, θ) of B are $\left(2, \frac{1}{6}\pi\right)$.

Given that A, B and O are points on a circle, find the radius of this circle.

Exercise 6

a(i). The equation of an ellipse E is given by

$$\frac{x^2}{9} + \frac{y^2}{4} = 1.$$

Show that the gradient of the tangent to E at the point $P(x, y)$ is $-\dfrac{4x}{9y}$.

a(ii). The equation of an ellipse E is $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$. A tangent is drawn to E from the point $Q(3, 4)$.

Show that the equation of one of the two tangents is $2y = x + 5$.

b. Solve the equation

$$7\sinh x - 5\cosh x + 1 = 0,$$

giving your answer in the form $\ln a$, where a is an integer to be determined.

Exercise 7

a. By using the Intermediate Value Theorem, show that the equation $x^3 - x^2 - x - 5 = 0$ has at least one real solution.

b. Prove by induction that $12^n + 2 \times 5^{n-1}$ is divisible by 7 for all integers $n \geq 1$.

Exercise 8

i. A sequence u_n is defined by

$$u_1 = \frac{5}{2}, \qquad u_{n+1} - 3 = \frac{nu_n - 3n}{2(n+1)}.$$

Find u_2.

ii. A sequence u_n is defined by

$$u_1 = \frac{5}{2}, \qquad u_{n+1} - 3 = \frac{nu_n - 3n}{2(n+1)}.$$

Given that $v_n = n(u_n - 3)$, $k \in \mathbb{R}$, prove that $v_{n+1} = kv_n$, and find v_1 and k.

iii. A sequence u_n is defined by $u_1 = \frac{5}{2}$ and $u_{n+1} - 3 = \frac{nu_n - 3n}{2(n+1)}$, and $v_n = n(u_n - 3)$.

Find v_n in terms of n.

iv. A sequence u_n is defined by $u_1 = \frac{5}{2}$ and $u_{n+1} - 3 = \frac{nu_n - 3n}{2(n+1)}$, and $v_n = n(u_n - 3)$.

Hence show that $\lim_{n \to \infty} u_n = 3$.

v. A sequence u_n is defined by $u_1 = \frac{5}{2}$ and $u_{n+1} - 3 = \frac{nu_n - 3n}{2(n+1)}$.

State the bounds of u_n.

Exercise 9

i. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Find the domain of f.

ii. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Find $\lim_{x \to +\infty} f(x)$.

iii. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Find $\lim_{x \to 2} f(x)$.

iv. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Study the variation of $f(x)$.

v. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Sketch the curve $y = f(x)$.

vi. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Another function g is defined by $g(x) = \dfrac{f(x)}{x-3}$.

Study the sign of g as x varies across $x = 3$.

vii. A function f is defined by

$$f(x) = \frac{\ln(x-1)}{x-2}.$$

Another function g is defined by $g(x) = \dfrac{f(x)}{x-3}$.

Sketch the curve $y = g(x)$.

Exercise 10

a. Using De Moivre's theorem or otherwise, solve the equation $z^2 = 1 + i$.

a(i). A transformation T is defined by $T : z \to w$, where

$$w = \frac{3z + 4i}{z - 1}, \qquad z \neq 1.$$

Find the invariant point of this transformation.

a(ii). A transformation T is defined by $T : z \to w$, where

$$w = \frac{3z + 4i}{z - 1}, \qquad z \neq 1.$$

Show that the image of the circle $|z| = 1$ is a line in the w-plane.

b(i). $\bar{z}$ is the conjugate of the complex number z and $z\bar{z} = |z|^2$. Given that $|z - 3i| = |z - i|$, show that $\bar{z} = z - 4i$ and deduce that $z = x - i$, $x \in \mathbb{R}$.

b(ii). $\bar{z}$ is the conjugate of the complex number z and $z\bar{z} = |z|^2$. Given that $|z - 3i| = |z - i|$, if $\arg(z) = \theta$, find the range of values of θ.

b(iii). $\bar{z}$ is the conjugate of the complex number z and $z\bar{z} = |z|^2$. Given that $|z - 3i| = |z - i|$, if $\arg(z) = \theta$, deduce that $z = 2\cot\theta + 2i$.

b(iv). $\bar{z}$ is the conjugate of the complex number z and $z\bar{z} = |z|^2$. Given that $|z - 3i| = |z - i|$ and that $\arg\left(\dfrac{z}{z - 4i}\right) = \alpha$, show that $z = 2\cot\left(\frac{1}{2}\alpha\right) + 2i$.

b(v). $\bar{z}$ is the conjugate of the complex number z and $z\bar{z} = |z|^2$. Given that $|z - 3i| = |z - i|$ and that $\arg\left(\dfrac{z}{z - 4i}\right) = \alpha$,

hence or otherwise, find the $(n - 1)$ solutions of the equation $(z - 4i)^n = z^n$.

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

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